

OceanEmbed (SIH PS 26066) — Platform Feature Reference

AI-Driven 3D Subsurface Ocean Temperature & Salinity Digital Twin • Comprehensive Technical Manual

Target Domain: MoES SIH 2026 PS 26066	Spatial Grid: 30°N to -40°S, 30°E to 120°E (Indian Ocean)	Depth Domain: 0m to 1000m (CTD Benchmark)
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1. Top Navigation Bar (TopNavbar)

Control / Component	Element Type	Technical Purpose & Functionality
Subsurface Emblem & Logo	Interactive Brand Link	Resets camera view back to default Indian Ocean overview coordinates.
MoES PS 26066 Badge	Status Chip	Confirms official Ministry of Earth Sciences compliance and mission mode.
Geospatial Search Input	Search Bar + Auto-Suggest	Searches any coordinate (Lat, Lon) or Indian Ocean sub-basin (e.g. Arabian Sea, Bay of Bengal).
3D Volumetric / Globe Switch	Primary View Toggle	Switches between 3D Global Earth digital twin and isolated 3D Ocean Block Slice.
Constellation Button (Satellite)	Modal Trigger	Opens Satellite Constellation Telemetry Modal to inspect 6 Earth Observation orbiters.
Analytics Button (BarChart3)	Modal Trigger	Opens AI Telemetry & Model Evaluation Modal displaying R ² , RMSE, MAE, and sub-basin accuracy.
Mission Info Button (Info)	Modal Trigger	Opens Mission Pipeline & Architecture Modal outlining 3-stage deep neural reconstruction workflow.
UTC System Clock & Status Pill	Live Indicator	Displays live synchronized UTC timestamp with green pulse confirming active telemetry feed.

2. Interactive 3D Digital Twin Globe (Main Canvas)

Control / Component	Element Type	Technical Purpose & Functionality
Photorealistic Earth Mesh	3D Interactive Model	High-res bathymetric topology with cloud layers and Rayleigh atmospheric scattering glow.
Satellite Constellation Orbits	Real-Time 3D Paths	Renders 6 live satellites orbiting along realistic orbital inclinations with trailing telemetry rings.
Indian Ocean Perimeter Loop	Glowing 3D Polyline	Outlines exact regional boundary (30°N to -40°S, 30°E to 120°E) with subtle breathing pulse.
Live Argo Float Array Pins	Interactive Markers	Represents in-situ CTD floats. Hovering displays telemetry; clicking loads vertical depth profile.
Subsurface Cutaway Cylinder	Volumetric Pillar	Renders physical 0–1000m thermal stratification at selected coordinate with laser depth rings.
Orbit Controls	Camera Controller	Smooth physics-damped orbit camera allowing full multi-angle inspection and zoom of ocean features.

3. Left Control Panel: Telemetry & Ingestion (LeftSidebar)

Control / Component	Element Type	Technical Purpose & Functionality
Basin & Coordinates Card	Dynamic Header	Displays active ocean location name, exact latitude/longitude, and nearest Argo WMO station ID.
SST Hero Card	Live Telemetry Display	Shows live Sea Surface Temperature (°C), AI Confidence Gauge (e.g., 96.4%), and timestamp.
Subsurface Profile Matrix	4-Metric Grid	Displays: Surface SST (0m), Thermocline Base (~200m), Abyssal Temp (1000m), and Mixed Layer Depth (MLD).
Sub-Basin Quick-Jump Chips	Filter Buttons	One-click teleportation to key regional zones: Arabian Sea, Bay of Bengal, Equatorial IO, Southern Ocean.
3D Ocean Block Button	Action Button (Layers)	Launches isolated 3D volumetric slice view for localized subsurface inspection.
Full Depth Profile Button	Action Button (Activity)	Opens complete 0–1000m vertical CTD temperature and salinity distribution chart.

4. Right Control Panel: Layers & Depth Scrubber (RightSidebar)

Control / Component	Element Type	Technical Purpose & Functionality
Observation Layer Buttons	Mode Selectors	Toggles geospatial heatmaps: Temperature (°C), Salinity (PSU), Surface Currents (m/s), Atmospheric Coupler.
Auto-Orbit Toggle Switch	Toggle Control	Enables or pauses automatic slow cinematic planetary rotation around the Indian Ocean basin.
Subsurface Depth Slider	Continuous Range Slider	Sweeps interactive depth plane from 0m (Surface) down to 1000m (Abyssal Floor) with live feedback.

Ocean Stratification Zone Pill	Contextual Badge	Auto-updates based on depth: Epipelagic Mixed Layer (0-100m), Thermocline (100-450m), Deep Ocean (450-1000m).
Quick Jump Depth Chips	Preset Buttons	One-tap navigation to standard oceanographic benchmarks: 0m, 50m, 150m, 300m, 500m, 1000m.

5. 3D Ocean Block Slice Modal (OceanBlockView)

Control / Component	Element Type	Technical Purpose & Functionality
Camera View Presets	Multi-Button Pill	Shifts 3D slice camera to: Isometric (3D view), Front (East-West cutaway), Side (North-South), Top (Surface).
Reset Orbit Button	Icon Button (RotateCcw)	Restores 3D block to default perspective view angle.
Subsurface Telemetry Card	Compact HUD Box	Displays exact slice temperature at current depth, ΔT gradient drop from surface, and AI confidence R².
Thermal Spectrum Legend	Color Map HUD	Compact visual gradient scale mapping colors from -2°C (Deep blue/violet) to >32°C (Warm crimson).
Depth Scrubber Plane Slider	Range Slider	Sweeps highlighted horizontal laser slice up and down through the 3D water block in real-time.
Auto Sweep Play/Pause	Animation Toggle	Automatically animates continuous vertical scanning from 0m to 1000m and back.
Stratification Quick Jumps	Preset Buttons	Jumps to standard layers: Surface (0m), Mixed Layer (75m), Thermocline (200m), Mesopelagic (500m), Abyssal (1000m).

6. Full CTD Depth Profile Modal (FullProfileModal)

Control / Component	Element Type	Technical Purpose & Functionality
Vertical Temperature Graph	Recharts Interactive Area	High-density curve showing temperature decline from surface down to 1000m with thermocline inflection points.
Physical Layer Indicators	Chart Annotations	Highlights Mixed Layer Depth (MLD), Main Thermocline gradient zone, and Deep Isothermal water.
Summary Metric Cards	3-Card Header	Displays Surface SST (°C), Thermocline Core Depth (m), and Abyssal Floor Temp (°C).
Download Profile Data Button	Export Action	Exports 0–1000m interpolated profile as formatted CSV / JSON data file for scientific reporting.

7. Satellite Constellation Modal (ConstellationModal)

Control / Component	Element Type	Technical Purpose & Functionality
Satellite Selection List	Left Master List	Displays active satellites: Sentinel-6 Michael Freilich, Jason-3, INSAT-3DR, Oceansat-3 (EOS-06), SWOT, SARAL/AltiKa.
Satellite Specification Matrix	Right Detail Card	Displays orbital altitude, inclination, payload sensor packages, revisit frequency, and subsystem health %.
Live Ground Track & Pass Time	Real-Time Metric	Shows last overpass timestamp over specific Indian Ocean sub-basins (e.g., Over Bay of Bengal 14 mins ago).

8. AI Telemetry & Model Analytics Modal (TelemetryAnalyticsModal)

Control / Component	Element Type	Technical Purpose & Functionality
Global Benchmark Metrics	4-KPI Overview	Displays RMSE (0.28°C), R² Score (0.945), Mean Absolute Error (0.21°C), and Inference Latency (<12ms).
Regional Sub-Basin Matrix	Comparison Grid	Side-by-side performance evaluation for Arabian Sea, Bay of Bengal, Equatorial Indian Ocean, and Southern Ocean.
Depth Strata Accuracy Bars	Stratification Breakdown	Visual bar breakdown validating AI reconstruction accuracy across Epipelagic, Thermocline, Intermediate, and Bathypelagic layers.

9. Mission Info & Architecture Modal (MissionInfoModal)

Control / Component	Element Type	Technical Purpose & Functionality
Problem Statement Overview	Scientific Abstract	Details MoES SIH 2026 Problem Statement 26066 objectives and spatial coverage boundaries.
3-Tier Pipeline Architecture	Visual Flow Cards	1. Surface Data Ingestion (SST, SSHa, Wind) -> 2. Deep Neural Subsurface Inversion -> 3. 3D WebGL Digital Twin.
Operational Impact Panel	Impact Badges	Explains downstream benefits for Cyclone Intensity Forecasting, Monsoon Dynamics, Fisheries, and Naval Operations.

10. Floating HUD Widgets & Bottom Status Bar

Control / Component	Element Type	Technical Purpose & Functionality
North Compass Widget	Top-Right Floating Dial	Visual orientation dial with cardinal points (N, S, E, W); clicking re-aligns view to true North.
Float Telemetry Card	Hover Floating HUD	Appears when hovering over any Argo pin, showing float WMO ID, live depth, temp, salinity, and last ping.
Mobile Drawer Dock	Bottom Floating Dock	Touch dock allowing mobile users to toggle between Sidebar Controls, Depth Slider, and Layer Switcher.
Bottom Status Bar	Bottom Live Strip	Operational status strip with INCOIS / MoES institutional badges, active float counts, and live system latency.